

VIVOWATCH OPEN API For NHRI

認證

VIVOWATCH Open API 使用 Bearer Token 来进行認證。在每 Request 的

Header 必須包含以下格式的

Authorization: Bearer <access_ticket>

Ocp-Apim-Subscription-Key: e3a74968f57c4d2d9a01a66bc46ee364

呼叫流程

首先呼叫 GetAuth 取得 token, 再使用 DataWithTimePeriod 指定時間取得資

料，token 的效期為取得後的 24 小時內有效，如果收到已過期的回應請再呼叫

GetAuth 取得 token.

無資料的兩種不同情況說明

有兩種不同的狀況會查無資料，以下分別說明傳回值的不同

傳回值	情況
http status: 204. Content length: 0	查詢時間內並沒有資料被上傳，最常見的原因是資料尚未同步到雲端。請

	檢查 APP 中 [備份資料]->最新備份檔案日期.
http status: 200，且有 deviceid 但沒有任何資料.如下: <pre>{ "daily_data": [{ "deviceid": "L8HCGX101284", "hb": [], "bp": [], "spo2": [], "step": [], "tp": [] }] }</pre>	代表查詢的區間內資料有從 APP 備份到雲端，但那個時間區間內並沒有任何資料。最常見的情況就是沒有量測到數值 另一個少見的情況是因為某些網路因素導致那個小時雖然有資料但查詢區間的資料尚未完全備份到雲端，這種情況再等各 5-10 分鐘應該就會有資料。

API 說明

API URL: <https://aocc-api-mgm.azure-api.net>

POST /vivowatch/bpapi/GetAuth	取得有效 1 天的 Token
Header: Authorization: 固定字串, Bearer bdae6eb7870ae752169f333863a2bfbb337ed44681b4dfff23adccae40eb58454 6ec988dc59c4ba7b91f32dabe6c01eb9eab47ad540157addb4fbb1d1edd4329 Ocp-Apim-Subscription-Key: 固定字串, e3a74968f57c4d2d9a01a66bc46ee364	

輸入參數:JSON 格式字串	
source_domain	固定字串: nhri.edu.tw
apiid	固定字串: nhri-auth-001
回傳值: JSON 格式字串	
token	從取得開始的 24 小時內有效的字串
HTTP Code	
200	成功
400	錯誤的參數 回傳 JSON 中的 status_code TOKEN_NOT_FOUND : HTTP 標題未帶入 token PARAMETERS_REQUIRED: 未帶入必要參數
401	token 錯誤
500	內部嚴重錯誤，請聯絡華碩人員

使用 curl 呼叫及傳回值範例

```
curl --location 'https://aocc-api-mgm.azure-api.net/vivowatch/bpapi/GetAuth' \
--header 'Ocp-Apim-Subscription-Key: ee6fad3f352643de829ee4535918640c' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
bdae6eb7870ae752169f333863a2bfbb337ed44681b4dfff23adccae40eb584546ec988dc59
c4ba7b91f32dabe6c01eb9eab47ad540157addb4fbb1d1edd4329' \
--data '{
    "source_domain":"nhri.edu.tw",
    "apiid":"nhri-auth-001"
}'
```

傳回值

```
{
  "token":"[token]"
}
```

POST /vivowatch_2b_data/v1/bpapi/DataWithTimePeriod 指定時間區間取回所有資料	
Authorization Header: Bearer <i>token</i> Ocp-Apim-Subscription-Key: 固定字串, e3a74968f57c4d2d9a01a66bc46ee364	填入從 GetAuth 所取得的 <i>token</i>
輸入參數: JSON 格式字串	
start_time	起始日期與時間(24h), 格式: yyyy-MM-dd HH:mm:ss
end_time	結束日期與時間(24h), 格式: yyyy-MM-dd HH:mm:ss
回傳值: JSON 格式字串	
daily_data (陣列)	日常的生理資料, 以裝置為單位為一個 JSON 物件
deviceid	裝置序號
bp	血壓 type: 1(手動), 129(背景量測) time: 時間 sys: 收縮壓 dia: 舒張壓 hr: 當次量測到的心跳 deStressIndex: 壓力指數 rmssd: 壓力指數的原始資料
hb	心跳 time: 時間 heartrate: 心跳數值
spo2	血氧 type: 0(手動量測), 1(背景量測) time: 時間
step	步數 time: 時間 steps: 步數
tp	體表溫度 time: 時間 temerature: 溫度

bia	身體組成資訊: water/fat/skm 三個數值都先除以 10 後才代表% time:時間 water: 水分 (三位數, 例如 584, 代表 58.4%) fat: 體脂(三位數, 例如 221, 代表 22.1%) skm: 骨骼肌肉(三位數, 例如 217, 代表 21.7%) weight: 體重
HTTP Code	
200	成功
400	錯誤的參數 回傳 JSON 中的 status_code TOKEN_NOT_FOUND : HTTP 標題未帶入 token PARAMETERS_REQUIRED: 未帶入必要參數
204	時間區間內無資料
401	token 錯誤或失效
500	內部嚴重錯誤, 請聯絡華碩人員

使用 curl 呼叫及傳回值範例

```
curl --location 'https://aocc-api-mgm.azure-api.net/vivowatch_2b_data/v1/bpapi/DataWithTimePeriod' \
--header 'Ocp-Apim-Subscription-Key: e3a74968f57c4d2d9a01a66bc46ee364' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
Rr8OnBGQlrqd8K1bwMqPCxZLfygljXdW5/Xrj5BoTsoYxDWSDsmo/v+/9UQ8IEPNwMp+plve
8rOA3hsTh7ysla+0QVO6jNF7hlabojlpc67OWZbkfNraYQq0bICSGKe3fQCeIQyZoX6kUmk8
N3VDQ==' \
--data '{
  "start_time":"2023-09-12 00:00:00",
  "end_time":"2023-09-14 11:30:00"
}'
```

傳回值範例請參考章節:資料範例中的生理資料

POST /vivowatch_2b_data/v1/bpapi/	
ExerciseDataWithTimePeriod 指定時間區間取回運動資料	
Authorization Header: Bearer <i>token</i> Ocp-Apim-Subscription-Key: 固定字串, e3a74968f57c4d2d9a01a66bc46ee364	填入從 GetAuth 所取得的 <i>token</i>
輸入參數: JSON 格式字串	
start_time	起始日期與時間(24h), 格式: yyyy-MM-dd HH:mm:ss
end_time	結束日期與時間(24h), 格式: yyyy-MM-dd HH:mm:ss
回傳值: JSON 格式字串	
(陣列)	運動資料
deviceid	裝置序號
type	運動類型 室內跑步: 1 室外跑步: 33 室內游泳: 2 室外游泳: 34 飛輪: 3 室外走路: 35 室外自行車: 36 瑜伽: 5 拉筋: 6 核心運動: 7 舞蹈: 8 健身: 9 室內腳踏車: 10
totalDistance	總距離
totalTime	總時間(分鐘)
totalStep	總步數
totalCalories	消耗的總熱量
exerciseDatetime	運動開始時間
latitudeArrayList	每兩秒的緯度資訊
longitudeArrayList	每兩秒的經度資訊
altitudeArrayList	每兩秒的高度資訊

speedArrayList	每兩秒一次的速度紀錄，單位是 0.01km/h
distanceArrayList	每兩秒一次的距離 累積 紀錄，單位是 m
heartRateArrayList	每兩秒一次的心率紀錄
aerobicArrayList	每兩秒一次的有氧紀錄 0：一般，1：有氧，2：無氧
stepArrayList	每兩秒一次的步數累積紀錄
caloriesArrayList	每兩秒一次的卡路里累積紀錄
skinTempArrayList	每兩秒一次的溫度紀錄. 例如：35560 代表 35.5. 無條件捨去後面的 60
HTTP Code	
200	成功
400	錯誤的參數 回傳 JSON 中的 status_code TOKEN_NOT_FOUND ：HTTP 標題未帶入 token PARAMETERS_REQUIRED ：未帶入必要參數
204	時間區間內無資料
401	token 錯誤或失效
500	內部嚴重錯誤，請聯絡華碩人員

使用 curl 呼叫及傳回值範例

```
curl --location 'https://aocc-api-mgm.azure-
api.net/vivowatch_2b_data/v1/bpapi/ExerciseDataWithTimePeriod' \
--header 'Ocp-Apim-Subscription-Key: e3a74968f57c4d2d9a01a66bc46ee364' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
9WbnxkDSm7aXdZ8vr5S0I5GyeJH9V3zqX7PMUUaMaS34RZloOyWhqfHet7mgiaFh5YbUvR
BSwvo7/MGboyCQ5LglBoAcuBzcrOY/u+2iqGzDJ81Qo65Fjm+XsDxajEl5UrE2pRxhifMzVuIB
TYB6DA==' \
--data '{
  "start_time":"2023-09-28 00:00:00",
  "end_time":"2023-09-30 00:00:00"
}'
```

傳回值範例請參考章節:資料範例中的運動資料

POST /vivowatch_2b_data/v1/bpapi/SleepDataWithTimePeriod 指定時間區間取回睡眠資料.	
Authorization Header: Bearer <i>token</i> Ocp-Apim-Subscription-Key: 固定字串, e3a74968f57c4d2d9a01a66bc46ee364	填入從 GetAuth 所取得的 <i>token</i>
輸入參數: JSON 格式字串，睡眠是以天為單位計算，只要輸入日期即可	
start_time	起始日期 (24h), 格式: yyyy-MM-dd
end_time	結束日期 (24h), 格式: yyyy-MM-dd
回傳值: JSON 格式字串	
(陣列)	睡眠資料
deviceid	裝置序號
StartTime	起始時間
EndTime	結束時間
Toss	翻身次數
ComfortCount	舒眠時間(分)
LightCount	淺眠時間(分)
AwakeCount	清醒時間(分)
RemCount	眼動時間(分)
HTTP Code	
200	成功
400	錯誤的參數 回傳 JSON 中的 status_code TOKEN_NOT_FOUND : HTTP 標題未帶入 token PARAMETERS_REQUIRED: 未帶入必要參數
204	時間區間內無資料
401	token 錯誤或失效
500	內部嚴重錯誤，請聯絡華碩人員

使用 curl 呼叫及傳回值範例

```
curl --location 'https://aocc-api-mgm.azure-  
api.net/vivowatch_2b_data/v1/bpapi/SleepDataWithTimePeriod' \  
--header 'Ocp-Apim-Subscription-Key: e3a74968f57c4d2d9a01a66bc46ee364' \  
--header 'Content-Type: application/json' \  
--header 'Authorization: Bearer  
9WbnxkDSm7aXdZ8vr5S0I5GyeJH9V3zqX7PMUUaMaS34RZloOyWhqfHet7mgiaFh5YbUvR  
BSwvo7/MGboyCQ5LglBoAcuBzcrOY/u+2iqGzDJ81Qo65Fjm+XsDxajEI5UrE2pRxhifMzVuIB  
TYB6DA==' \  
--data '{  
    "start_time":"2023-09-17",  
    "end_time":"2023-09-18"  
}'
```

傳回值範例請參考章節:資料範例中的運動資料

POST	
/vivowatch_2b_data/v1/bpapi/SleepDetailDataWithTimePeriod	
指定時間區間取回睡眠及其相關資料.	
Authorization Header: Bearer <i>token</i>	填入從 GetAuth 所取得的 <i>token</i>
Ocp-Apim-Subscription-Key: 固定字串, e3a74968f57c4d2d9a01a66bc46ee364	
輸入參數: JSON 格式字串，睡眠是以天為單位計算，只要輸入日期即可	
start_time	起始日期 (24h), 格式: yyyy-MM-dd
end_time	結束日期 (24h), 格式: yyyy-MM-dd
deviceid	(選填、非必要)裝置序號
回傳值: JSON 格式字串	
(陣列)	睡眠資料

deviceid	裝置序號
StartTime	起始時間
EndTime	結束時間
Toss	翻身次數
ComfortCount	舒眠時間(分)
LightCount	淺眠時間(分)
AwakeCount	清醒時間(分)
RemCount	眼動時間(分)
ActivityState(陣列)	每一分鐘的睡眠狀態 數值解釋： 8 代表 REM 9 代表 Light sleep 10 代表 comfort sleep 其餘數值都算做清醒
HeartRate(陣列)	每一分鐘的心跳數值。但因為背景心跳量測是依照使用者自己設定而且有時會量不到，所以沒有量到的數值就為 0
Spo2(陣列)	每一分鐘的血氧數值。但因為背景血氧量測是依照使用者自己設定而且有時會量不到，所以沒有量到的數值就為 0。超過 128 的數值直接減去 128
HTTP Code	
200	成功
400	錯誤的參數 回傳 JSON 中的 status_code TOKEN_NOT_FOUND : HTTP 標題未帶入 token PARAMETERS_REQUIRED: 未帶入必要參數
204	時間區間內無資料
401	token 錯誤或失效
500	內部嚴重錯誤，請聯絡華碩人員

使用 curl 呼叫及傳回值範例

```
curl --location 'https://aocc-api-mgm.azure-api.net/vivowatch_2b_data/v1/bpapi/SleepDetailDataWithTimePeriod' \
--header 'Ocp-Apim-Subscription-Key: e3a74968f57c4d2d9a01a66bc46ee364' \
--header 'Content-Type: application/json' \
```

```
--header 'Authorization: Bearer
9WbnxkDSm7aXdZ8vr5S0I5GyeJH9V3zqX7PMUUaMaS34RZloOyWhqfHet7mgiaFh5YbUvR
BSwvo7/MGboyCQ5LglBoAcuBzcrOY/u+2iqGzDJ81Qo65Fjm+XsDxajEI5UrE2pRxhifMzVuIB
TYB6DA==' \
--data '{
  "start_time":"2023-08-26",
  "end_time":"2023-08-27"
}'
```

傳回值範例請參考章節:資料範例中的睡眠及其相關資料

資料範例

1. 生理資料

```
{ "daily_data": [ { "deviceid": "R5HCGX700722CZ3", "hb": [ { "time": "2023-09-12 12:24:00",
"heartrate": 92 }, { "time": "2023-09-12 12:35:00", "heartrate": 45 }, { "time": "2023-
09-12 12:46:00", "heartrate": 84 }, { "time": "2023-09-12 12:57:00", "heartrate": 77 },
{ "time": "2023-09-12 13:01:00", "heartrate": 78 }, { "time": "2023-09-12 13:13:00",
"heartrate": 109 }, { "time": "2023-09-12 13:19:00", "heartrate": 98 }, { "time":
"2023-09-12 13:32:00", "heartrate": 120 }, { "time": "2023-09-12 13:43:00", "heartrate":
75 }, { "time": "2023-09-12 13:45:00", "heartrate": 85 }, { "time": "2023-09-12
13:55:00", "heartrate": 80 }, { "time": "2023-09-12 13:58:00", "heartrate": 76 } ],
"bp": [ { "type": 129, "time": "2023-09-12 12:46:00", "sys": 121, "dia": 81, "hr": 84,
"rmssd": 25, "deStressIndex": 80 }, { "type": 129, "time": "2023-09-12 13:01:00", "sys":
118, "dia": 78, "hr": 78, "rmssd": 31, "deStressIndex": 85 }, { "type": 129, "time":
"2023-09-12 13:19:00", "sys": 123, "dia": 80, "hr": 98, "rmssd": 24, "deStressIndex": 79 },
{ "type": 129, "time": "2023-09-12 13:45:00", "sys": 123, "dia": 81, "hr": 85, "rmssd": 30,
"deStressIndex": 85 }, { "type": 129, "time": "2023-09-12 13:58:00", "sys": 115, "dia": 75,
"hr": 76, "rmssd": 39, "deStressIndex": 91 } ], "spo2": [], "step": [ { "time": "2023-09-12
11:57:00", "steps": 6 }, { "time": "2023-09-12 11:58:00", "steps": 6 }, { "time":
"2023-09-12 12:20:00", "steps": 95 }, { "time": "2023-09-12 12:21:00", "steps": 115 },
{ "time": "2023-09-12 12:22:00", "steps": 54 }, { "time": "2023-09-12 12:23:00", "steps":
62 }, { "time": "2023-09-12 12:24:00", "steps": 11 }, { "time": "2023-09-12 12:36:00",
"steps": 49 }, { "time": "2023-09-12 12:37:00", "steps": 89 }, { "time": "2023-09-12
12:38:00", "steps": 50 }, { "time": "2023-09-12 12:39:00", "steps": 114 }, { "time":
```

```
"2023-09-12 12:40:00", "steps": 101 }, { "time": "2023-09-12 12:41:00", "steps": 38 },
{ "time": "2023-09-12 12:44:00", "steps": 32 }, { "time": "2023-09-12 12:52:00", "steps":
27 }, { "time": "2023-09-12 13:21:00", "steps": 33 }, { "time": "2023-09-12 13:24:00",
"steps": 33 }, { "time": "2023-09-12 13:25:00", "steps": 52 }, { "time": "2023-09-12
13:29:00", "steps": 85 }, { "time": "2023-09-12 13:30:00", "steps": 89 }, { "time":
"2023-09-12 13:31:00", "steps": 102 }, { "time": "2023-09-12 13:32:00", "steps": 12 },
{ "time": "2023-09-12 13:33:00", "steps": 42 }, { "time": "2023-09-12 13:34:00", "steps":
31 }, { "time": "2023-09-12 13:35:00", "steps": 36 }, { "time": "2023-09-12 13:36:00",
"steps": 23 }, { "time": "2023-09-12 14:07:00", "steps": 13 } ], "tp": [] },
{ "deviceid": "N7HCGX500252KMB", "hb": [ { "time": "2023-09-13 08:41:00", "heartrate":
73 }, { "time": "2023-09-13 08:53:00", "heartrate": 97 }, { "time": "2023-09-13
09:00:00", "heartrate": 100 }, { "time": "2023-09-13 09:05:00", "heartrate": 100 },
{ "time": "2023-09-13 09:06:00", "heartrate": 96 }, { "time": "2023-09-13 09:07:00",
"heartrate": 97 }, { "time": "2023-09-13 09:08:00", "heartrate": 90 }, { "time": "2023-
09-13 09:09:00", "heartrate": 104 }, { "time": "2023-09-13 09:10:00", "heartrate": 110 },
{ "time": "2023-09-13 09:21:00", "heartrate": 83 }, { "time": "2023-09-13 09:39:00",
"heartrate": 92 }, { "time": "2023-09-13 09:50:00", "heartrate": 82 }, { "time": "2023-
09-13 10:01:00", "heartrate": 85 }, { "time": "2023-09-13 10:02:00", "heartrate": 88 },
{ "time": "2023-09-13 10:12:00", "heartrate": 86 }, { "time": "2023-09-13 10:23:00",
"heartrate": 90 }, { "time": "2023-09-13 10:36:00", "heartrate": 93 }, { "time": "2023-
09-13 10:46:00", "heartrate": 93 } ], "bp": [ { "type": 129, "time": "2023-09-13 09:00:00",
"sys": 140, "dia": 91, "hr": 100, "rmssd": 20, "deStressIndex": 74 }, { "type": 129,
"time": "2023-09-13 10:02:00", "sys": 131, "dia": 85, "hr": 88, "rmssd": 27, "deStressIndex":
82 } ], "spo2": [ { "type": 1, "time": "2023-09-13 09:17:00", "spo2": 99 },
{ "type": 1, "time": "2023-09-13 09:46:00", "spo2": 99 }, { "type": 1, "time": "2023-09-
13 10:07:00", "spo2": 96 }, { "type": 1, "time": "2023-09-13 10:28:00", "spo2": 98 } ],
"step": [ { "time": "2023-09-13 08:42:00", "steps": 79 }, { "time": "2023-09-13 08:43:00",
"steps": 27 }, { "time": "2023-09-13 08:58:00", "steps": 41 }, { "time": "2023-09-13
08:59:00", "steps": 23 }, { "time": "2023-09-13 09:05:00", "steps": 36 }, { "time":
"2023-09-13 09:06:00", "steps": 1 }, { "time": "2023-09-13 09:09:00", "steps": 57 },
{ "time": "2023-09-13 09:10:00", "steps": 109 }, { "time": "2023-09-13 09:11:00", "steps":
112 }, { "time": "2023-09-13 09:12:00", "steps": 44 }, { "time": "2023-09-13
09:14:00", "steps": 61 }, { "time": "2023-09-13 09:22:00", "steps": 43 }, { "time":
"2023-09-13 09:34:00", "steps": 9 }, { "time": "2023-09-13 09:41:00", "steps": 9 },
{ "time": "2023-09-13 09:42:00", "steps": 5 }, { "time": "2023-09-13 09:45:00", "steps":
33 }, { "time": "2023-09-13 09:57:00", "steps": 36 }, { "time": "2023-09-13 09:58:00",
"steps": 15 }, { "time": "2023-09-13 10:14:00", "steps": 38 }, { "time": "2023-09-13
10:15:00", "steps": 17 }, { "time": "2023-09-13 10:16:00", "steps": 74 } ], "tp":
```

[illegible]

2. 運動資料

[illegible]

1	],	"type": 35,	"heartRateArrayList": [	0,	0,	0,	0,	0,	
0,	0,	0,	0,	0,	0,	0,	0,	90,	90,
90,	90,	93,	94,	95,	98,	100,	102,	103,	
104,	104,	104,	104,	104,	104,	104,	104,	104,	104,
104,	104,	104,	104,	104,	104,	104,	104,	104,	104,
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3. 睡眠資料

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4. 睡眠及其相關資料

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